

Chapter 4

Elasticity

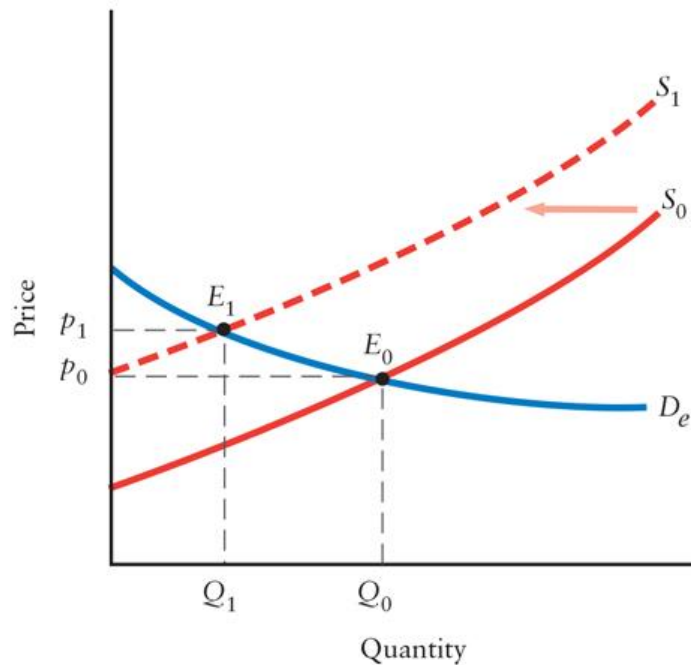
Chapter Outline/Learning Objectives

Section	Learning Objectives
	After studying this chapter, you will be able to
4.1 Price Elasticity of Demand	1. explain and calculate price elasticity of demand. 2. explain the relationship between total expenditure and price elasticity of demand.
4.2 Price Elasticity of Supply	3. explain and calculate price elasticity of supply.
4.3 Elasticity Matters for Excise Taxes	4. see how elasticity of demand and supply determine the effects of an excise tax.
4.4 Other Demand Elasticities	5. calculate the income elasticity of demand and distinguish between normal and inferior goods. 6. measure cross elasticity of demand and distinguish between substitute and complement goods.

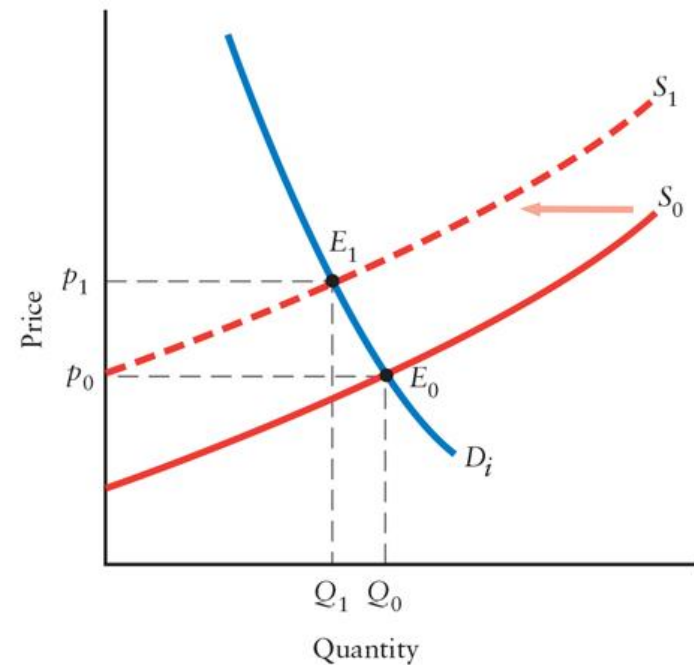
4.1 Price Elasticity of Demand

- Demand is *elastic* when quantity demanded is quite responsive to changes in price.
- When quantity demanded is relatively unresponsive to changes in price, demand is *inelastic*.
- The more elastic is demand, the less the change in equilibrium price and the greater the change in equilibrium quantity resulting from any given shift in the supply curve.

Figure 4-1 The Effects of a Supply Shift with Two Different Demand Curves



(i) Elastic demand



(ii) Inelastic demand

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The Measurement of Price Elasticity (1 of 2)

Price Elasticity of Demand (Greek letter eta: η) is defined as:

$$\eta = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$\eta = \frac{\frac{DQ_d}{\bar{Q}_d}}{\frac{DP}{\bar{P}}}$$

where P is the average price and Q_d is the average quantity.

The Measurement of Price Elasticity (2 of 2)

- A demand curve has a negative slope, so the percentage changes in price and quantity have opposite signs.
- Although demand elasticity is a negative number we ignore the negative sign and report the elasticity of demand as a positive number.

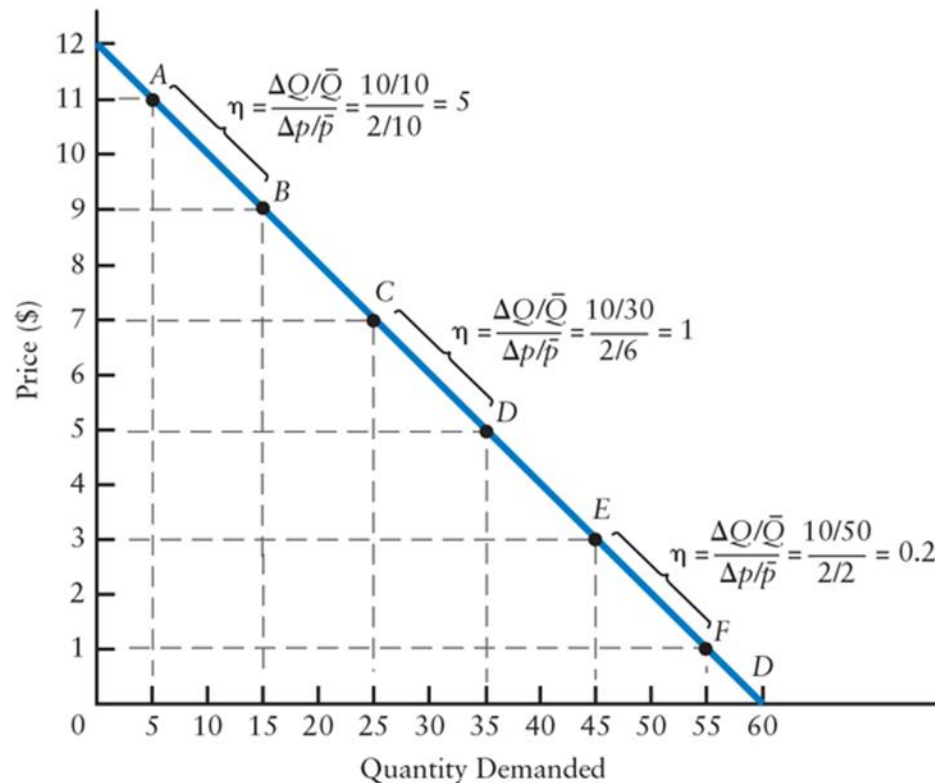
A Numerical Example of Price Elasticity

Product	Original Price	New Price	Average Price	Original Quantity	New Quantity	Average Quantity
Cheese	\$5.00	\$3.00	\$4.00	116 250	123 750	120 000

$$\eta = \frac{\frac{(123\,750 - 116\,250)}{120\,000}}{\frac{(5.00 - 3.00)}{4.00}} = \frac{7500 / 120\,000}{2.0 / 4.0} = \frac{0.0625}{0.5} = 0.125$$

Figure 4-2 Elasticity Along a Linear Demand Curve

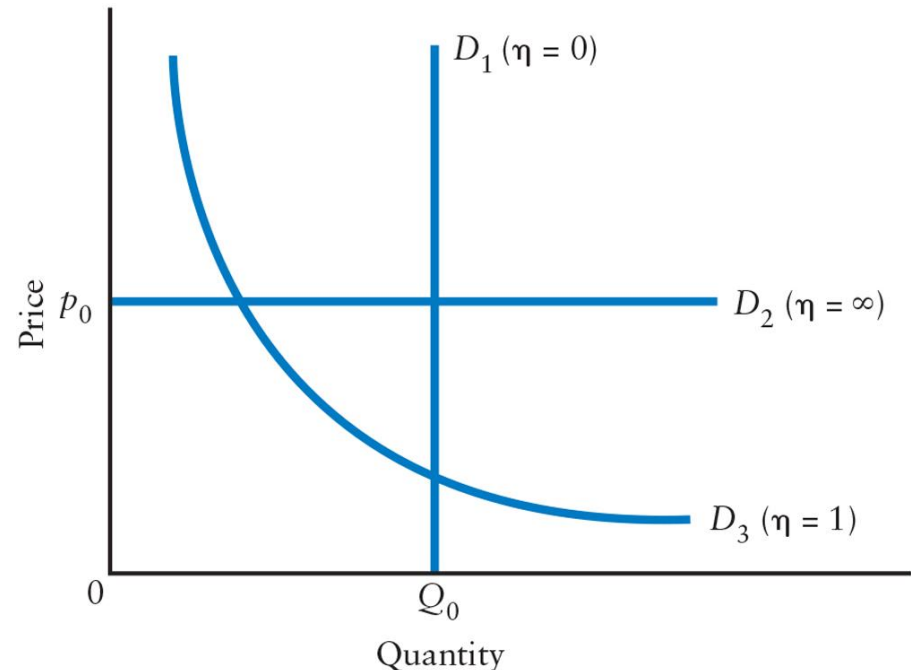
- A negatively sloped linear demand curve has a constant slope but does not have constant elasticity.



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Figure 4-3 Three Demand Curves with Constant Elasticity

- **D_1** shows **perfectly inelastic** demand
- **D_2** shows **perfectly elastic** demand
- **D_3** shows **unit elastic** demand



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What Determines Elasticity of Demand?

(1 of 3)

- **Availability of Substitutes**
 - Products with close substitutes tend to have elastic demands.
 - Products with no close substitutes tend to have inelastic demands.
 - Narrowly defined products have more elastic demands than do more broadly defined products.

What Determines Elasticity of Demand?

(2 of 3)

- **Importance of the Product in Consumers' Budgets**
 - Products that represent a small fraction of consumers' budgets tend to have inelastic demand.
 - Products that represent a large fraction of consumers' budgets tend to have elastic demand.
 - The general result is higher price elasticity for more expensive items.

What Determines Elasticity of Demand?

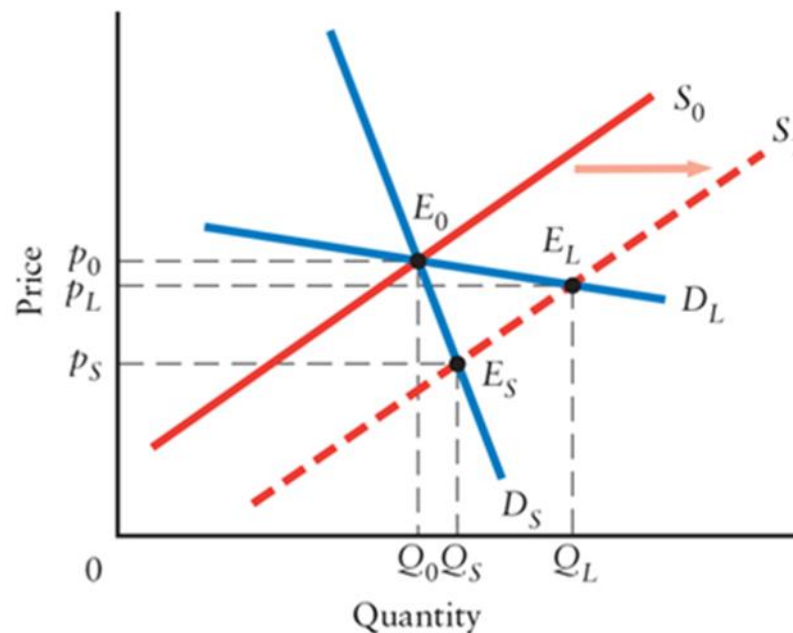
(3 of 3)

- **Short Run and Long Run**

- The longer the time span, the more elastic is the price elasticity of demand.
 - A ***short-run demand curve*** shows the immediate response of quantity demanded to a change in price.
 - A ***long-run demand curve*** shows the response of quantity demanded to a change in price after enough time has passed to develop or switch to substitute products.

Figure 4-4 Short-Run and Long-Run Equilibrium Following an Increase in Supply

- The changes depend on the time that consumers have to respond.
- In the long run, demand is more elastic.



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Elasticity and Total Expenditure

- Total expenditure = Price $\times$ Quantity
- How does total expenditure change when the price changes?
 - When price falls, quantity demanded increases.
 - When price increases, quantity demanded decreases.
- The change in total expenditure depends on price elasticity of demand.

Elasticity and Total Expenditure When Price Falls

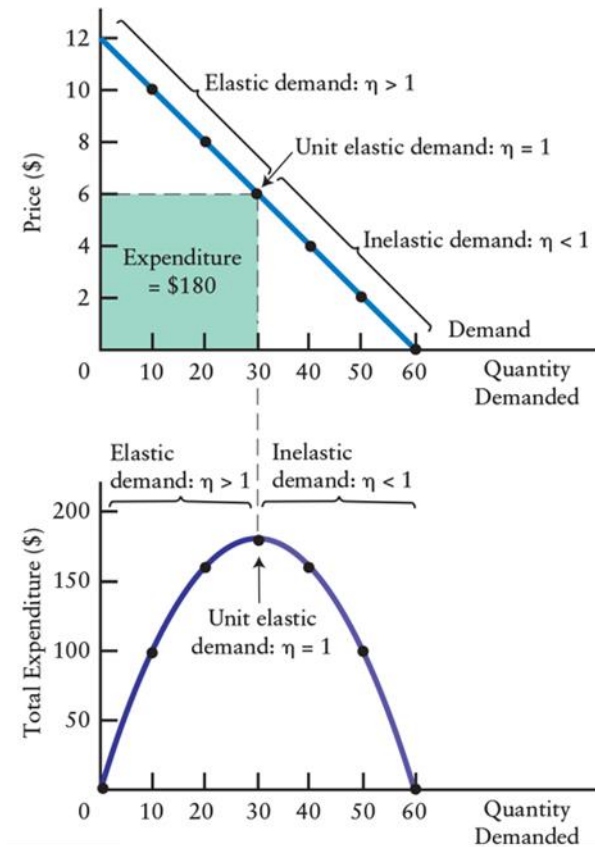
- If demand is **elastic**, the quantity change dominates and total expenditure rises.
- If demand is **inelastic**, the price change dominates and total expenditure falls.
- If demand is **unit elastic**, the percentage change in quantity demanded equals the percentage change in price and total expenditure remains unchanged.

Elasticity and Total Expenditure When Price Increases

- If demand is **elastic**, the quantity change dominates and total expenditure decreases.
- If demand is **inelastic**, the price change dominates and total expenditure increases.
- If demand is **unit elastic**, the percentage change in quantity demanded equals the percentage change in price and total expenditure remains unchanged.

Figure 4-5 Total Expenditure and Quantity Demanded

Price (\$)	Quantity Demanded	Expenditure (\$)
12	0	0
10	10	100
8	20	160
6	30	180
4	40	160
2	50	100
0	60	0



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4.2 Price Elasticity of Supply (1 of 2)

- Price elasticity of supply is a measure of the responsiveness of quantity supplied to a change in the product's own price.
- Supply is *elastic* when quantity supplied is quite responsive to changes in price.
- When quantity supplied is relatively unresponsive to changes in price, supply is *inelastic*.

4.2 Price Elasticity of Supply (2 of 2)

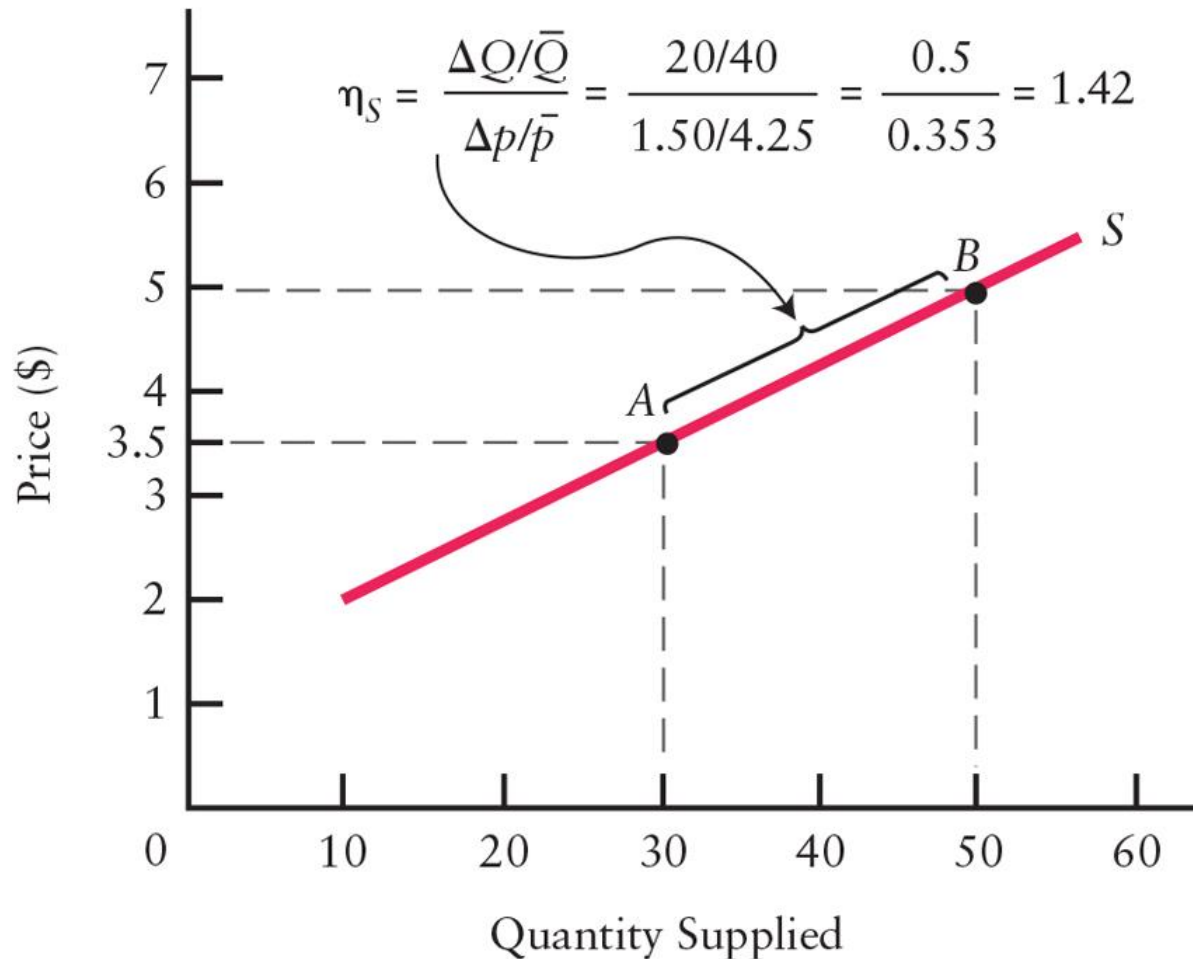
- **Price Elasticity of Supply** is denoted by η_s and is defined as:

$$\eta_s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$\eta_s = \frac{\frac{DQ_s}{\bar{Q}_s}}{\frac{DP}{\bar{P}}}$$

where P is the average price and Q_s is the average quantity.

Figure 4-6 Computing Price Elasticity of Supply



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Determinants of Supply Elasticity (1 of 2)

- **Ease of Substitution**

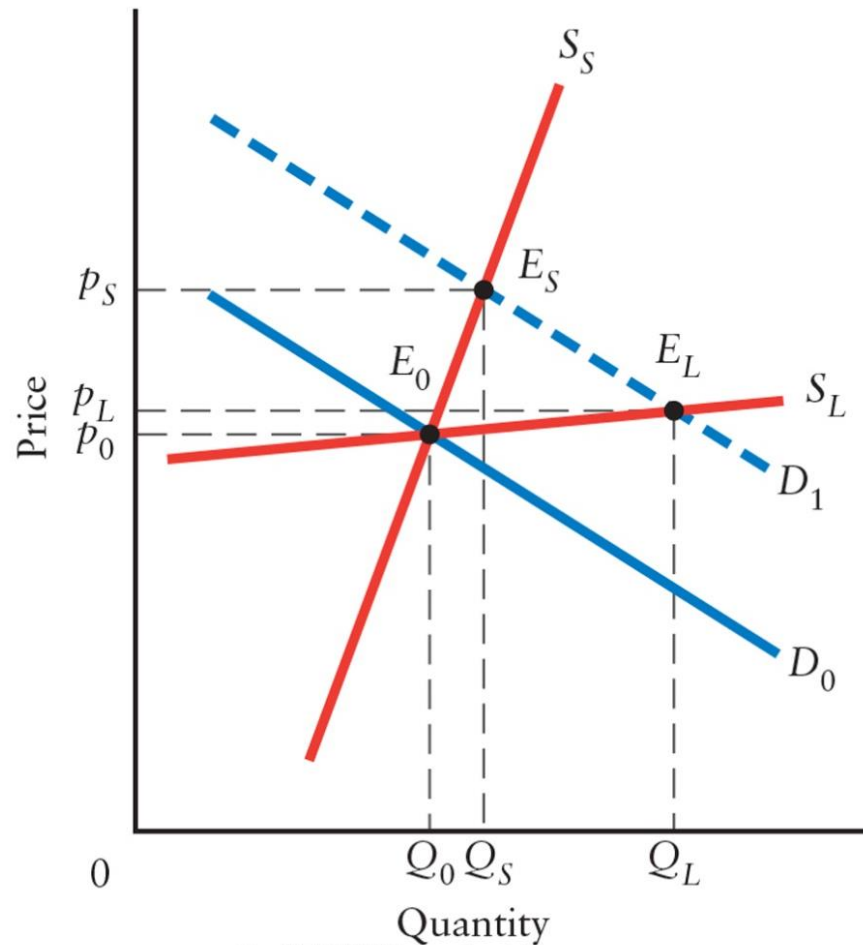
- If the price of a product rises, how much more can be produced profitably depends on how easy it is for producers to shift from the production of other products to the one whose price has risen.
- The easier it is to shift from the other products to the one whose price has risen, the more elastic the supply is.

Determinants of Supply Elasticity (2 of 2)

- **Short Run and Long Run**

- The *short-run supply curve* shows the immediate response of quantity supplied to a change in price given producers' current capacity to produce the good.
- The *long-run supply curve* shows the response of quantity supplied to a change in price after enough time has passed to allow producers to adjust their productive capacity.
- The shorter the time span is the harder it is to substitute away from one production and get into another.

Figure 4-7 Short-Run and Long-Run Equilibrium Following an Increase in Demand



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4.3 Elasticity Matters for Excise Taxes

(1 of 2)

- An **excise tax** is a tax on the sale of a particular product.
- An excise tax raises the price paid by consumers but reduces the price received by producers.

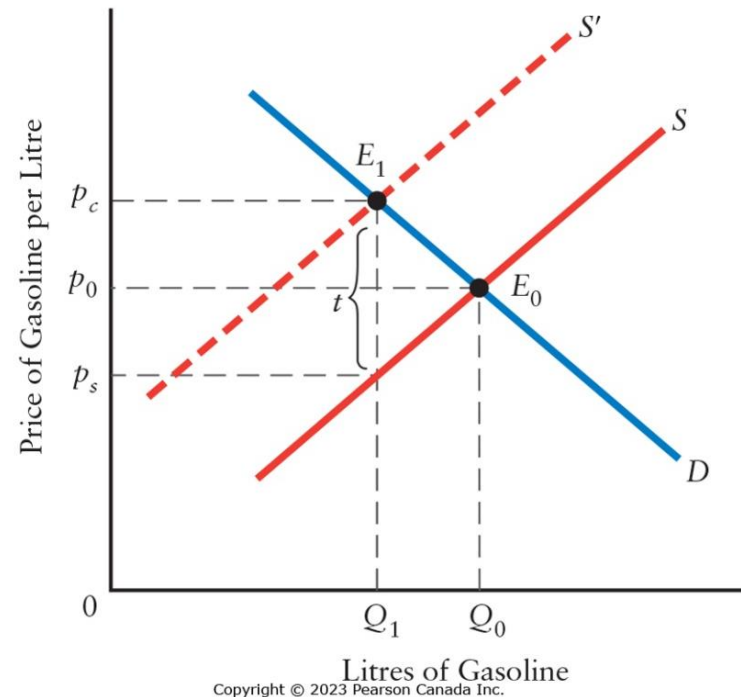


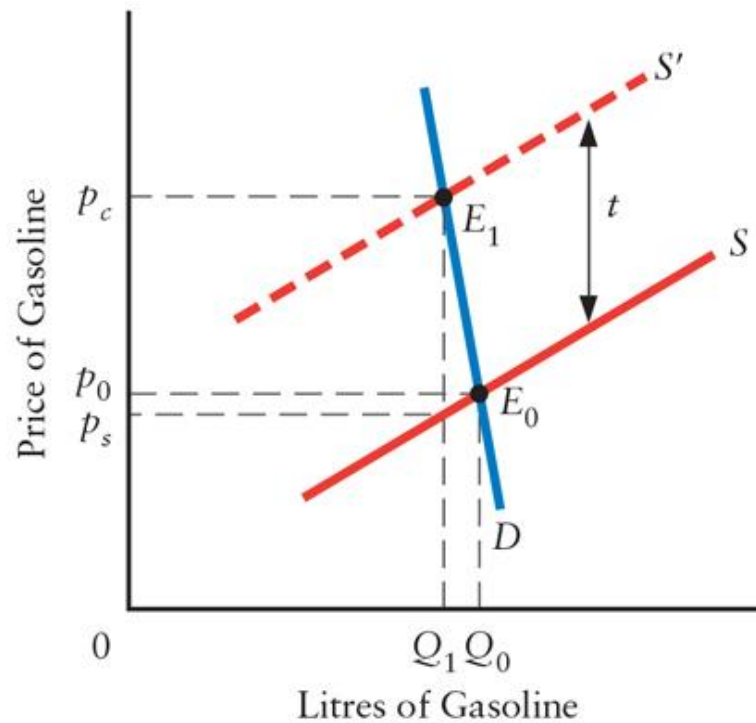
Figure 4-8 The Effect of a Gasoline Excise Tax

4.3 Elasticity Matters for Excise Taxes

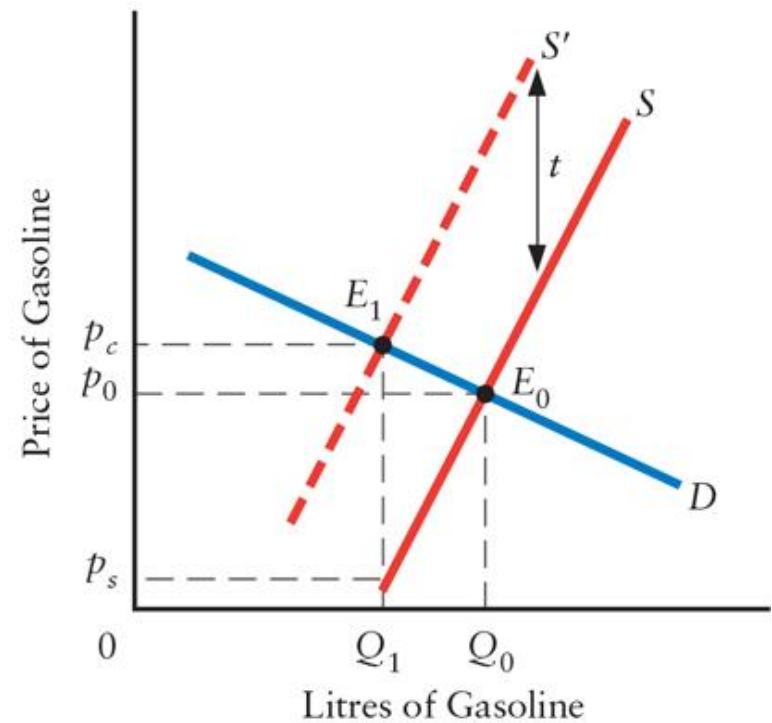
(2 of 2)

- Who actually bears the burden of this tax?
- The question of who bears *the burden* of a tax is called the question of **tax incidence**.
- With an excise tax, the difference between the price paid by the consumer and the price received by the seller is the amount of the tax.
- The burden of an excise tax depends on the relative elasticities of supply and demand.

Figure 4-9 Elasticity and the Incidence of an Excise Tax



(i) Inelastic demand, elastic supply



(ii) Inelastic supply, elastic demand

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Applying Economic Concepts 4-1

- **Who Really “Pays” for Payroll Taxes?**
 - Some social programs in Canada are financed by *payroll taxes* (Employment Insurance (EI) and Canada Pension Plan (CPP)).
 - When government proposes to increase EI or CPP premiums, the burden of extra taxes must be shared between workers and firms.
 - The allocation of this burden depends on the elasticities of labor demand and supply.
 - The more inelastic the labour supply is, the higher is the burden of the payroll tax on workers.
 - More elastic supply of and less elastic demand for labour may distribute the burden more equally.

4.4 Other Demand Elasticities

- **Income Elasticity of Demand**

$$\eta_Y = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

- If $\eta_Y > 0$, the good is a **normal** good.
- If $\eta_Y < 0$, the good is an **inferior** good.

Normal Goods

- If income elasticity is positive but less than one, demand is income inelastic.
- If income elasticity is positive and greater than one, demand is income elastic.
- Products for which the income elasticity of demand is positive but less than 1 are **necessities**.
- Products for which the income elasticity of demand is positive and greater than 1 are **luxuries**.
- The more necessary an item is in the consumption pattern of consumers, the lower is its income elasticity.

Inferior Goods

- Inferior goods have a negative income elasticity because an increase in income actually leads to a reduction in quantity demanded.
- It is generally easier to find examples of inferior goods for an individual than it is for the market as a whole.

Cross Price Elasticity of Demand (1 of 2)

- If $\eta_{XY} > 0$, then **X** and **Y** are **substitutes**.
- If $\eta_{XY} < 0$, then **X** and **Y** are **complements**.

$$\eta_{XY} = \frac{\text{Percentage change in quantity demanded of good X}}{\text{Percentage change in price of good Y}}$$

Cross Price Elasticity of Demand (2 of 2)

- The change in the price of good Y causes the demand curve for good X to shift.
- If X and Y are substitutes, an increase in the price of Y leads to an increase in the demand for X . The demand curve for X shifts rightward.
- If X and Y are complements, an increase in the price of Y leads to a decrease in the demand for X . The demand curve for X shifts leftward.